

# Deep Learning

# What is AI???



AI is the process of training machines to **imitate human intelligence and behavior** such as learning, reasoning, and decision-making.



Examples of Artificial Intelligence



Virtual Assistant



Face Recognition



Self-Driving Cars

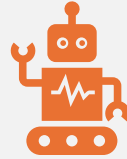


Chatbots



E-commerce Recommendations

# How is AI Achieved?



AI can be implemented through two key approaches:

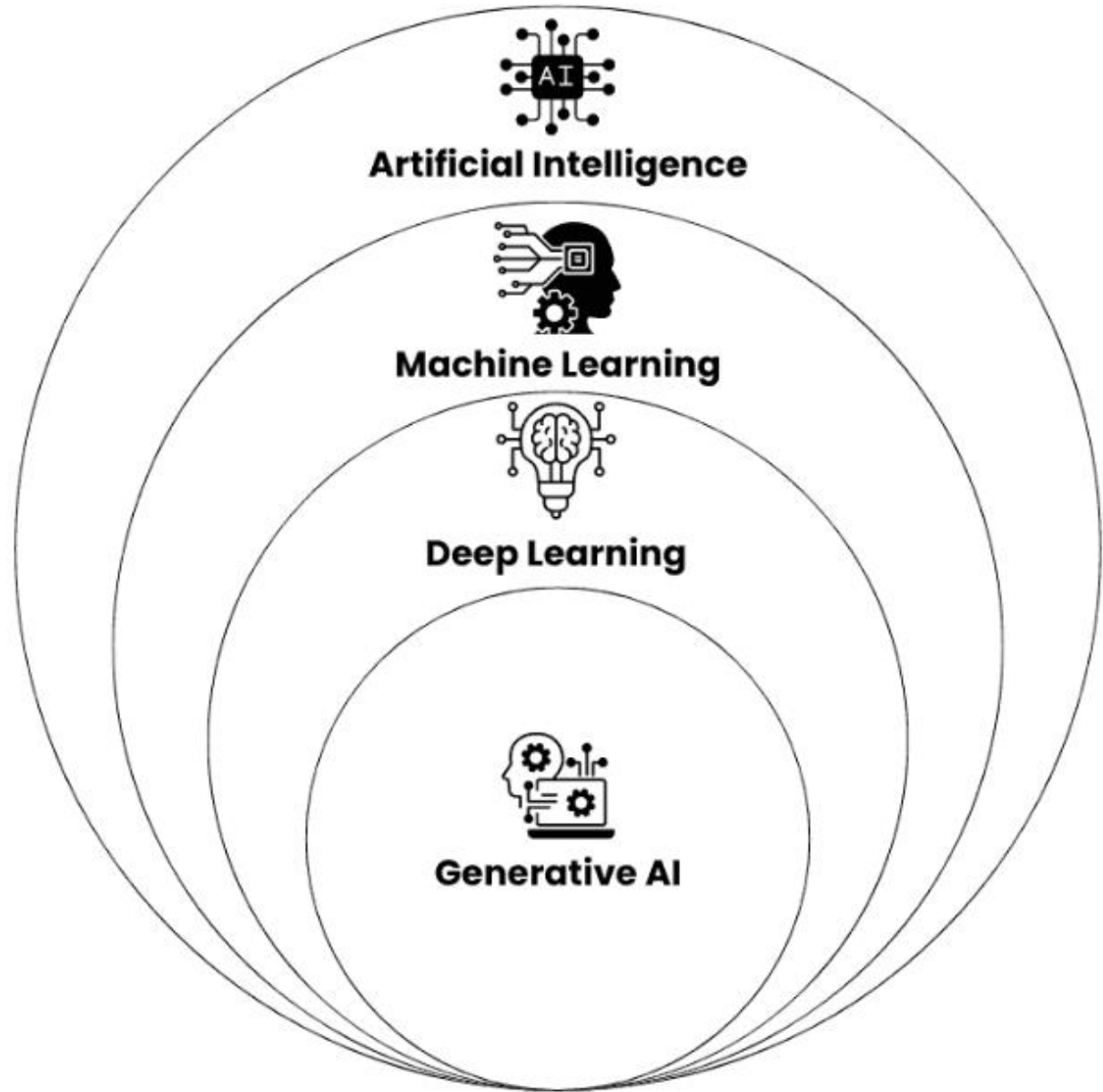


Machine Learning



Deep Learning

What is  
Deep  
learning ???





## Difference between ML and DL

### MACHINE LEARNING

Learn patterns from input data using algorithms such as SVM, LR, RF, and DT.

Uses shallow models

Manual feature engineering

Works on smaller training data

Less resource intensive

Easy to interpret

### DEEP LEARNING

Learn patterns using neural network.

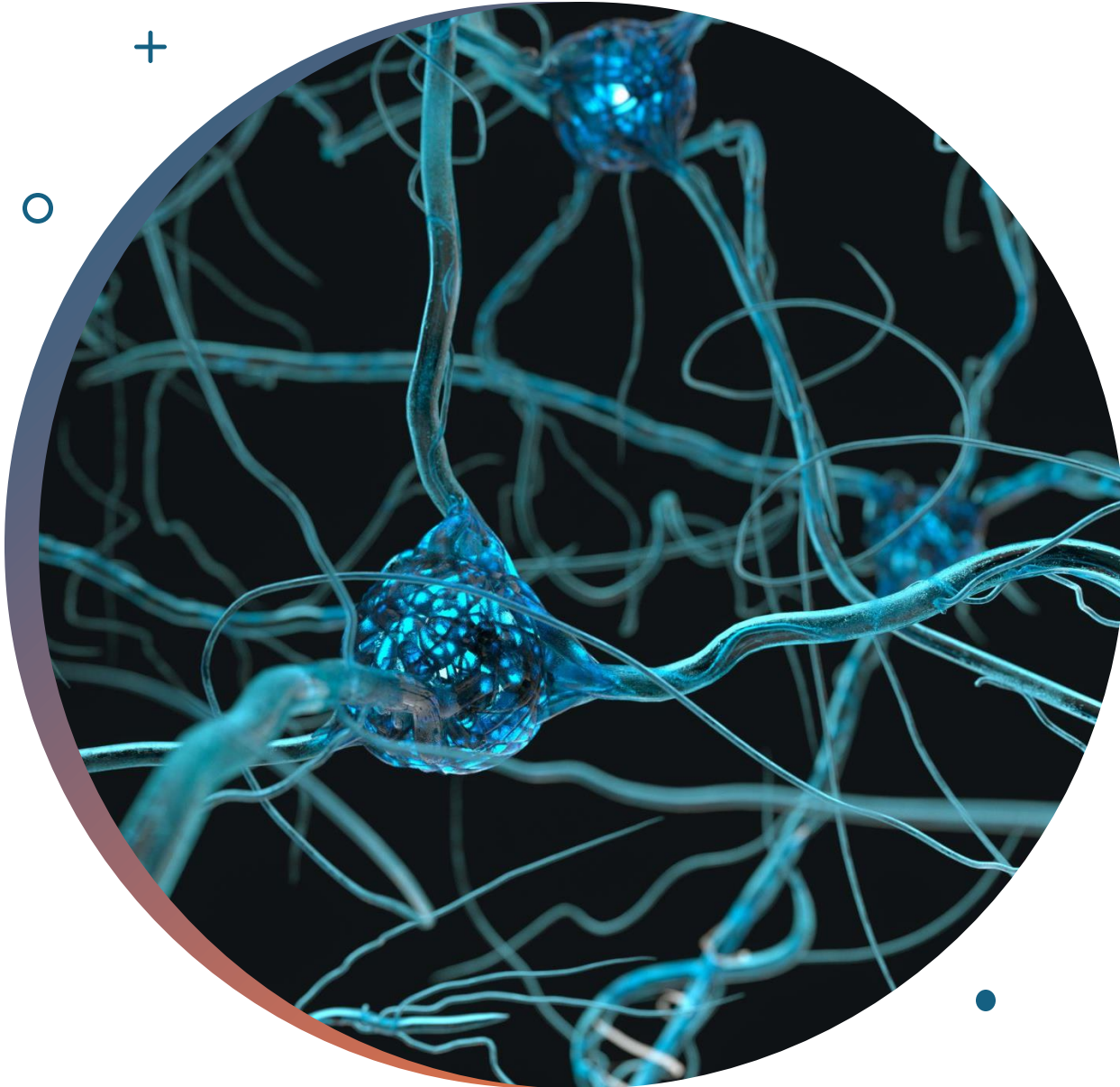
Uses Deep or more complex models

Automatically extract features

Requires large training corpus

Requires more computing power

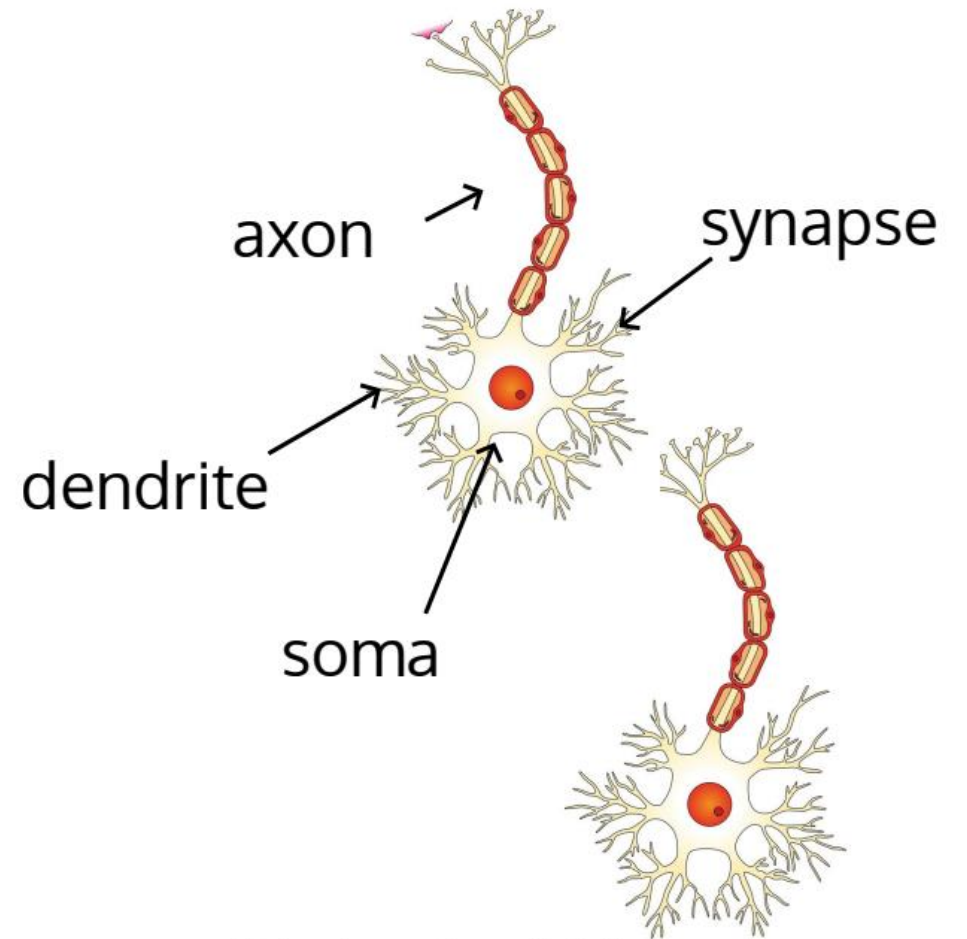
"Black box" difficult to interpret



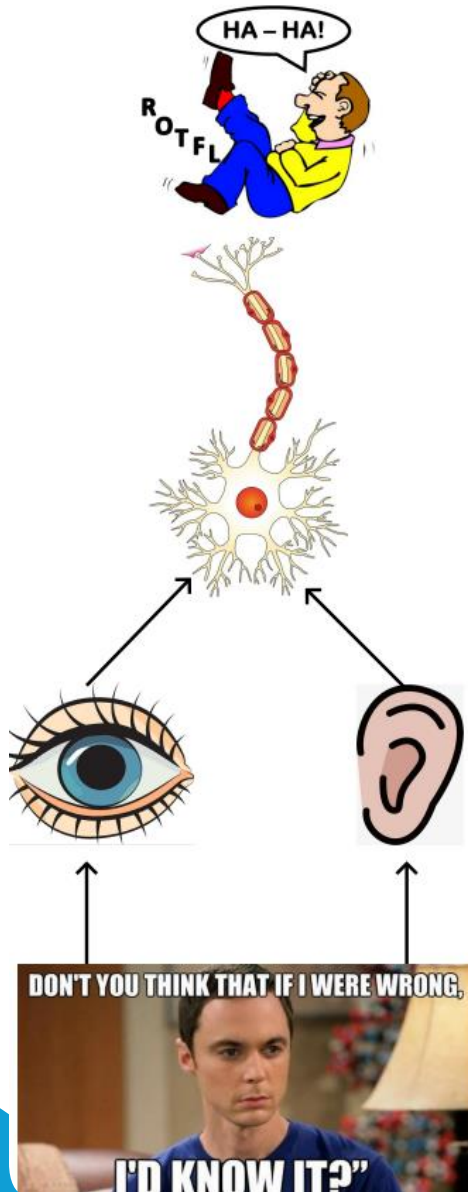
# Biological Neurons

- The most fundamental unit of a deep neural network is called an artificial neuron
- Why is it called a neuron ? Where does the inspiration come from ?
- The inspiration comes from biology (more specifically, from the brain)

- dendrite: receives signals from other neurons
- synapse: point of connection to other neurons
- soma: processes the information
- axon: transmits the output of this neuro

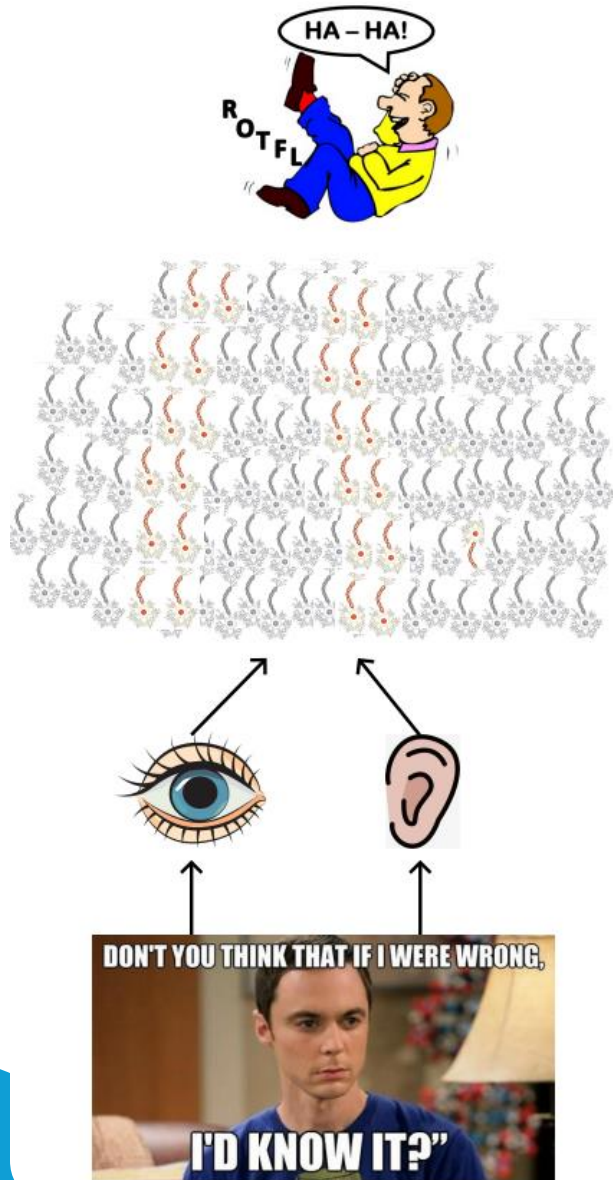


Biological Neurons\*



- Let us see a very cartoonish illustration of how a neuron works
- Our sense organs interact with the outside world
- They relay information to the neurons
- The neurons (may) get activated and produces a response (laughter in this case)



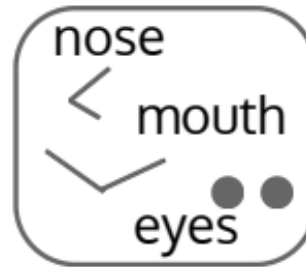


- Not just a single neuron which does all this
- There is a massively parallel interconnected network of neurons
- The sense organs relay information to the lowest layer of neurons
- Some of these neurons may fire (in red) in response to this information and in turn relay information to other neurons they are connected to
- An average human brain has around (100 billion) neurons!
- This massively parallel network also ensures that there is division of work
- Each neuron may perform a certain role or respond to a certain stimulus

- 
- The neurons in the brain are arranged in a hierarchy



Layer 1: detect edges & corners



Layer 2: form feature groups

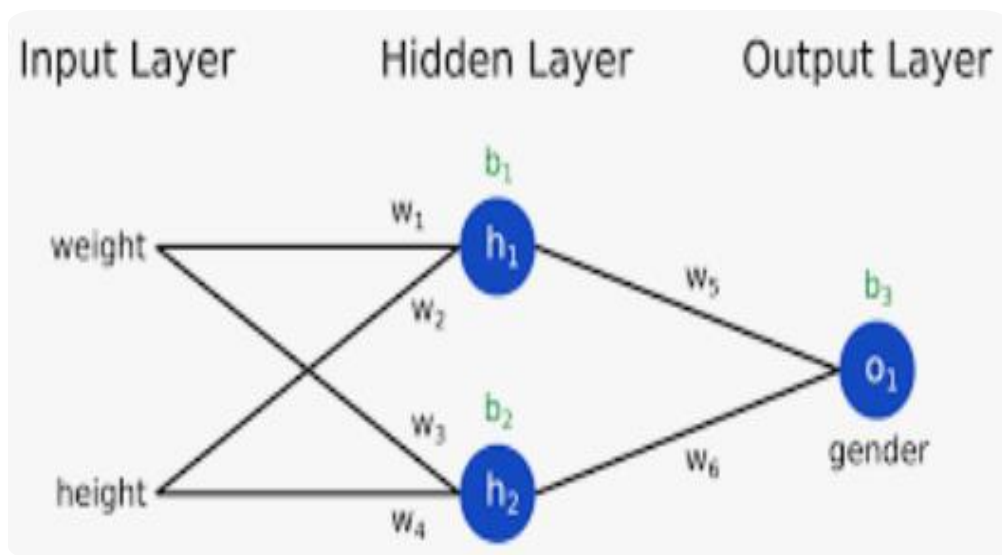


Layer 3: detect high level objects, faces, etc.

# Artificial neural network

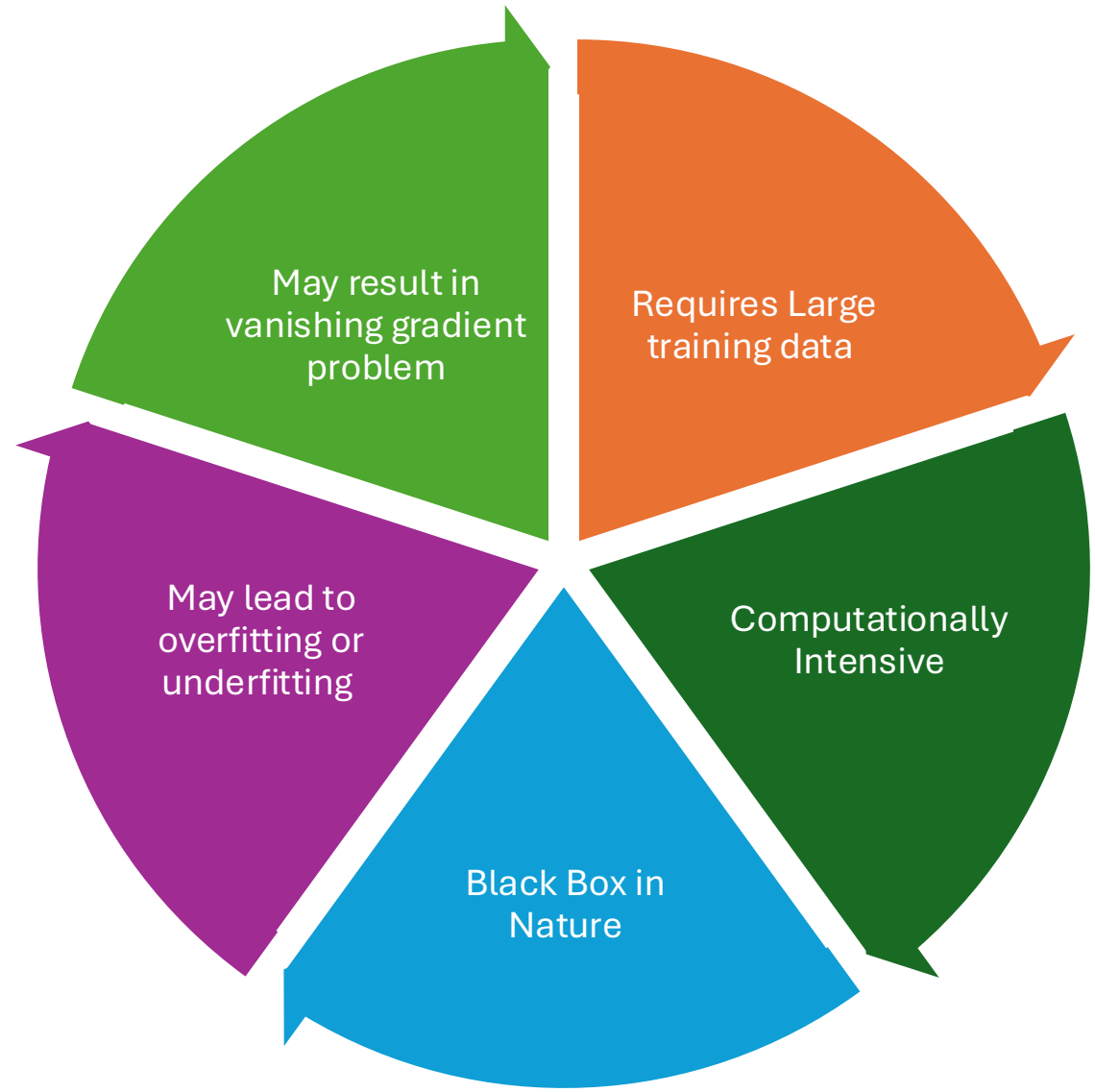
- Inspired by biological neuron in 1943 neurophysiologist Warren McCulloch Modeled a simple neural network using electrical circuits
- In 1950 first hypothetical neural network was simulated using computer by IBM research laboratory
- First attempt to build neuron was failed because of resource constraints
- In 1982, Paul Werbos applied backpropagation to neural networks. Later, it was popularized by Geoffrey Hinton and others, laying the groundwork for modern neural networks.

# Components of a Neural Network



- Input Layer
- Weights
- Bias
- Hidden Layers
- Output Layer

# Limitations of neural network



# Trends in Deep learning



Focus on model scale and efficiency :



Foundation Models & Large Language Models- Gemini, GPT-4o, Claude



Trained on massive data set with billions of parameters



Multimodal Learning- Combines **text + image + audio + video**



Smaller Language Models (SLMs)



Optimized Computing & Infrastructure

A large orange circle is positioned on the left side of the slide, partially cut off by the edge.

# Learning with less data

- Self-Supervised Learning
- Few-Shot and Zero-Shot Learning
- Transfer Learning



# Responsible AI development



Explainable AI



Bias and Fairness



Privacy Concerns



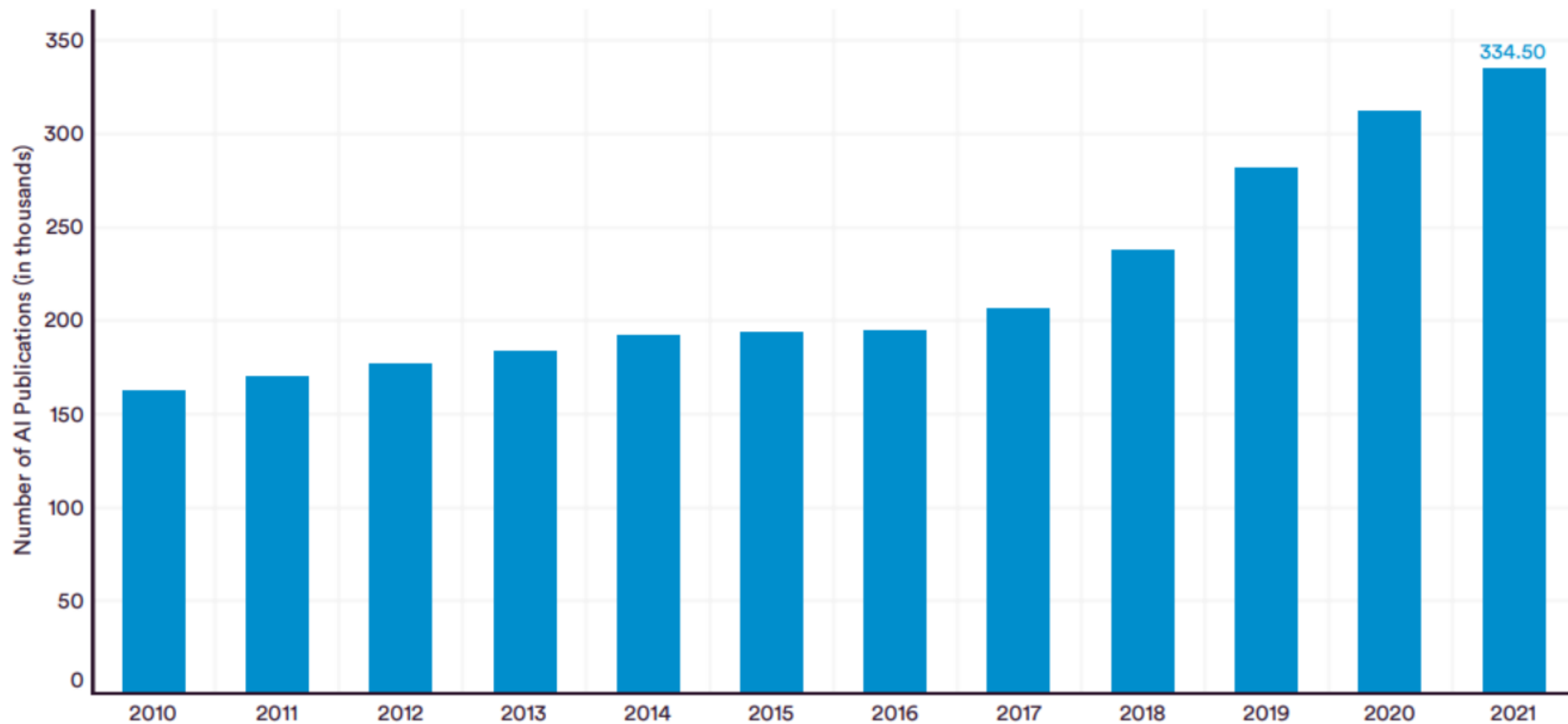
Ethical AI and Regulation



# AI Index Report 2022 (by Stanford)

## NUMBER of AI PUBLICATIONS in the WORLD, 2010–21

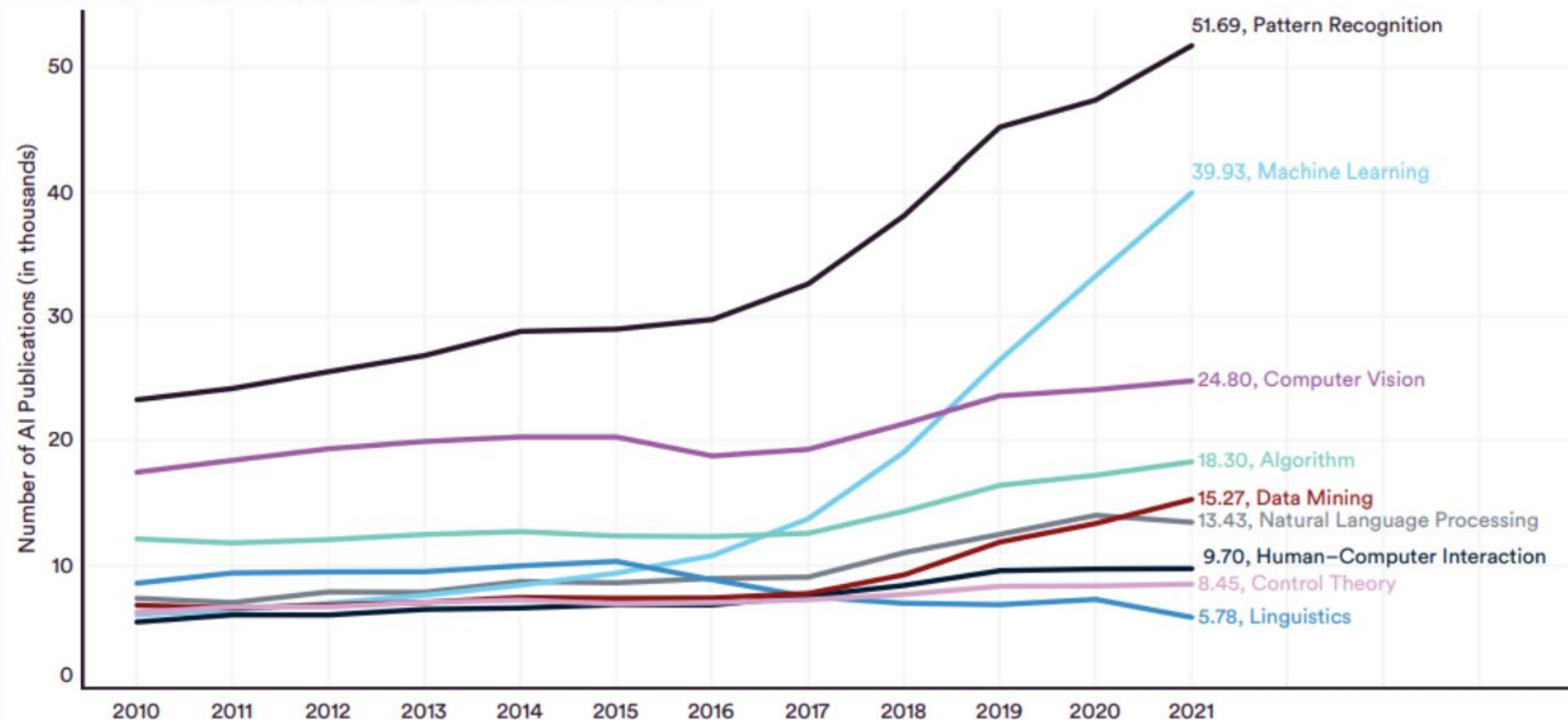
Source: Center for Security and Emerging Technology, 2021 | Chart: 2022 AI Index Report



# AI Index Report 2022 (by Stanford)

## NUMBER of AI PUBLICATIONS by FIELD of STUDY (EXCLUDING OTHER AI), 2010–21

Source: Center for Security and Emerging Technology, 2021 | Chart: 2022 AI Index Report



# AI Index Report 2022 (by Stanford)

## NUMBER of AI PATENT FILINGS, 2010–21

Source: Center for Security and Emerging Technology, 2021 | Chart: 2022 AI Index Report

